BIKA WUNGUKE — NISR data-fit review and clearer competition framing

Main recommendation

Your concept is relevant to NISR's 2026 Big Data Hackathon, especially **Track 2: Financial Inclusion & Poverty Reduction**. The strongest version is not a general finance app for everyone. Focus it on **small and informal traders who need a simple way to understand their business cash flow and prepare a clear financial record**. NISR's data can demonstrate the scale and context of the problem; the app should use each user's own entries to create that person's cash-flow record. The competition permits NISR data together with other open or public data and asks for a usable, data-informed solution. 1

What the concept already says—and what needs sharpening

The note proposes a mobile application for small businesses and a broad range of other users. It would record income and expenses, track savings, produce reports, provide AI advice, assess loan repayment capacity, and help users prepare information for lenders. The note also correctly says that the app cannot guarantee that a user will receive a loan and that financial reports should only be shared with consent.

The main issue is scope. Traders, farmers, motorcycle-taxi operators, artisans, young entrepreneurs, and households have different income patterns and record-keeping needs. Trying to serve all of them in the first prototype may make it harder to show a clear problem, data method, and impact. For the hackathon, lead with one user: **a self-employed micro-trader or informal small-business owner**. Other groups can be future users.

The other important adjustment is to separate **financial readiness** from **loan approval or credit scoring**. The concept can help a user keep records and understand whether cash is left after costs. But the NISR surveys are not a live record of that user's daily sales, expenses, debts, or repayment history. They therefore cannot, by themselves, tell the app whether a particular individual can safely repay a new loan.

How the NISR data fits

FinScope 2024 is the best dataset for the financial-inclusion evidence. It covers access to and use of financial services, savings, borrowing, and financial needs. NISR's catalogue describes it as a survey of eligible household members aged 16 or older, with one person

interviewed per selected household. The public-use microdata file is listed as 13,994 cases and 2,500 variables. The catalogue identifies it as the FinScope 2024 study and attributes production to Access to Finance Rwanda, so in your submission describe it accurately as **FinScope 2024, listed in NISR's Central Data Catalog and produced by Access to Finance Rwanda**. [2] [3]

FinScope supports the case for helping self-employed people prepare better records: its 2024 findings report that 42% of self-employed people who had obtained credit borrowed mainly to expand or invest in their businesses. This is evidence of business-related borrowing among borrowers; it is not evidence that all small traders are excluded or that keeping an app ledger will secure a loan. [2]

IBES 2024 is the best source for the business context. NISR reports that at least 87% of Rwanda's business landscape was informal in 2024. The publication also identifies insufficient collateral as the leading finance-access barrier, reported for 19.6% of informal businesses and 11.4% of formal businesses. These figures give your proposal a concrete national context, but they do not prove that poor bookkeeping is the cause of the collateral problem. Frame your app as helping traders produce clearer records—not as solving collateral requirements on its own. [6]

EICV7 is optional supporting context. It is NISR's 2023–2024 Integrated Household Living Conditions Survey, with national coverage and information used to monitor household living conditions and poverty. It could help you discuss poverty or vulnerability context, but it is not necessary for a focused prototype about trader cash-flow records. [7]

The FinScope microdata are listed as public-use files, but the NISR catalogue requires users to log in or register before access. Check the questionnaire and variable descriptions before finalizing analysis, and follow the catalogue's data-use terms. Do not assume that every variable you want is available just because the dataset has many fields. [3] [4] [5]

Recommended data and prototype plan

Use **FinScope 2024** to describe patterns in financial-service use, savings, credit, and business-related financial needs among relevant groups, especially self-employed respondents. Use the survey's weights and design information where applicable, and clearly label results as population-level survey findings rather than predictions about an individual user. The FinScope sample is stratified and clustered, so simple unweighted percentages may misrepresent the population. [2]

Use **IBES 2024 published results** to introduce the scale of informality and the reported financing barrier. You may add EICV7 only if you have a specific poverty-related question that the other sources cannot answer. [6] [7]

For the app demonstration, collect **synthetic demo entries or voluntary entries from consenting test users**: daily sales or income, business expenses, savings contributions, and dates. Generate a weekly or monthly summary showing total income, expenses, net cash flow, and progress toward a savings goal. Add an exportable report that the user can review and choose whether to share. Make it obvious which content comes from NISR statistics and which comes from the individual user’ s own entries.

Keep AI modest and auditable in the first version. A transparent summary or rule-based prompt—such as flagging a week when recorded expenses exceed recorded income—is easier to explain and validate than an AI-generated loan-readiness score. Do not label the output as a lender decision, a guaranteed repayment assessment, or a promise of credit access. Before any real lender integration, the project would need appropriate permission, privacy and security safeguards, validation, and a financial-institution partner.

Sharper project framing

Suggested title: *Bika Wunguke: a cash-flow and financial-readiness tool for small traders.*

One-sentence pitch: Bika Wunguke helps small traders record business income and expenses, understand their cash flow, build a savings habit, and prepare a user-controlled financial summary, while NISR data shows the wider financial-inclusion and informal-business context.

Suggested research question: What financial-service and borrowing patterns among self-employed people are visible in FinScope 2024, and how can a simple cash-book prototype help small traders organize the information they need to understand and explain their business cash flow?

Suggested objectives:

1. Use FinScope 2024 and IBES 2024 to describe relevant financial-inclusion and small-business financing patterns.
2. Build a simple prototype that records user-entered income, expenses, and savings and produces an understandable cash-flow summary.
3. Test whether intended users can enter transactions and understand the report in a short usability test.

Avoid claiming that the prototype will reduce poverty, increase loan approvals, or improve repayment unless you conduct a study that measures those outcomes. For a hackathon prototype, demonstrate usability and accurate reporting, and present broader effects as goals for future evaluation.

Suggested revised core text in Kinyarwanda

Ikibazo umushinga ugamije gukemura: Abacuruzi bato benshi, cyane cyane abakora mu rwego rutanditse, ntibafite uburyo bworoshye bwo kwandika amafaranga binjiza, ayo bakoresha n’ ayo babika. Ibyo bishobora kubagora kumenya uko ubucuruzi bwabo buhagaze no gutegura raporo y’ amafaranga binjiza n’ ayo basohora. Imibare ya NISR igaragaza ko ubucuruzi bwinshi mu Rwanda buri mu rwego rutanditse, kandi ko kutagira ingwate ihagije ari imwe mu mbogamizi zikomeye zo kubona imari. 6

Igisubizo: Bika Wunguke ni porogaramu izafasha umucuruzi muto kwandika amafaranga yinjira n’ asohoka, kureba amafaranga asigara, no gushyiraho intego yo kuzigama. Izajya ikora raporo y’ igihe runaka ishingiye ku makuru umukoresha yanditse, kandi uyikoresha ni we uzahitamo niba ayisangiza ikigo cy’ imari. Iyi porogaramu ntizemeza ko umuntu azahabwa inguzanyo kandi ntizafatira ikigo cy’ imari icyemezo ku bushobozi bw’ umukoresha bwo kwishyura.

Uko amakuru ya NISR azakoreshwa: Amakuru ya FinScope 2024 azakoreshwa mu kugaragaza uko abantu bikorera babona kandi bagakoresha serivisi z’ imari, harimo kwizigama no gufata inguzanyo. Ibyavuye muri IBES 2024 bizakoreshwa mu gusobanura uko ubucuruzi butanditse buhagaze n’ imbogamizi buhura na zo mu kubona imari. Aya makuru rusange ntazasimbura amakuru yihariye umukoresha azajya yandika muri porogaramu.

Fit with the 2026 competition

The official competition brief explicitly includes **Financial Inclusion & Poverty Reduction** and invites data-informed solutions with practical impact. It expects a public GitHub link, a deployed app link, and documentation. It also states that entrants transfer intellectual-property rights in submitted work to NISR and must disclose AI use. Read those terms before submitting. As of **16 October 2026**, the listed registration deadline is **20 October 2026 at 11:59 p.m.**, and the submission deadline is **30 October 2026 at 11:59 p.m.** 1

Overall assessment: Good thematic fit; strong public evidence for the problem context; incomplete fit if presented as an AI-based individual loan-capacity evaluator. Narrow the first version to trader cash-flow recording and user-controlled reporting. Use the NISR datasets to establish and analyze the wider need, not to imply that survey data can score a specific person.

References

[1] NISR 2026 Hackathon Competition
[2] Rwanda - FinScope 2024, Financial Inclusion in Rwanda: Study Description
[3] FinScope 2024 Data Dictionary
[4] FinScope 2024 Get Microdata

[5] FinScope 2024 Related Materials and Questionnaire
[6] NISR Integrated Business Enterprise Survey (2024)
[7] Rwanda - Integrated Household Living Conditions Survey (EICV7), 2023-2024